C6.4 How are chemicals made on an industrial scale? *(separate science only)*		
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*	***Linked learning opportunities***
Nitrogen, phosphorus and potassium are essential plant nutrient elements; they are lost from the soil when crops use them for growth and then are harvested. Fertilisers are added to the soil to replace these essential elements. The world demand for food cannot be met without the use of synthetic fertilisers. Natural fertilisers are not available in large enough quantities, their supply is difficult to manage and transport and their composition is variable. However, fertilisers can cause environmental harm when overused; if they are washed into rivers they cause excessive weed growth, which can lead to the death of the organisms that live there. Ammonia is one of the most important compounds used to make synthetic fertilisers. Ammonia is made in the Haber process, which involves a reversible reaction. **To get the greatest output as quickly and economically as possible chemical engineers consider the rate and the position of equilibrium for the reaction. In practice, industrial processes rarely reach equilibrium. In the Haber process unreacted reactants are continuously separated from the ammonia and recycled so that the nitrogen and hydrogen are not wasted.** Industrial processes need to be as economically profitable as possible. Atom economy is an indicator of the amount of useful product that is made in a reaction. This is a theoretical value based on the reaction equation and is used alongside data about yields and efficiency when processes are evaluated.	1. recall the importance of nitrogen, phosphorus and potassium compounds in agricultural production 2. explain the importance of the Haber process in agricultural production and the benefits and costs of making and using fertilisers, including: a) the balance between demand and supply of food worldwide b) the sustainability and practical issues of producing and using synthetic and natural fertilisers on a large scale c) the environmental impact of over-use of synthetic fertilisers (eutrophication) **3. explain how the commercially used conditions for the Haber process are related to the availability and cost of raw materials and energy supplies, control of equilibrium position and rate including:** **a) the sourcing of raw materials and production of the feedstocks; nitrogen (from air), and hydrogen (from natural gas and steam)** **b) the effect of a catalyst, temperature and pressure on the yield and rate of reaction** **c) the separation of the ammonia and recycling of unreacted nitrogen and hydrogen** **4. explain the trade-off between rate of production of a desired product and position of equilibrium in some industrially important processes** 5. define the atom economy of a reaction	**Specification links** • Haber Process (C4.3) **Practical Work:** • Laboratory preparation and purification of salts that are used as fertilisers. **Ideas about Science:** • Make predictions from data and graphs about yield of chemical products. (IaS2) • Consider the risks and costs of different operating conditions in an ammonia plant. (IaS4) • Production of fertilisers to enhance the quality of people's lives (IaS4) • Evaluation of industrial processes in terms of sustainability, risk, costs and benefits. (IaS4)

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Modern processes incorporate 'green chemistry' ideas, to provide a sustainable approach to production. Sustainability is a measure of how a process is able to meet current demand without having a long term impact on the environment. Reactions with high atom economy are more sustainable. Other issues which affect the sustainability of a process include; whether or not the raw materials are renewable; the impact of other competing uses for the same raw materials; the nature and amount of by-products or wastes; the energy inputs or outputs. Synthetic fertilisers contain salts that are made in acid-base reactions and can be synthesised on a laboratory scale. Scaling up of fertiliser manufacture for industrial production uses some similar processes to the laboratory preparation, but these are adapted to handle the much larger quantities involved. Any one compound used in fertilisers can often be made using several different processes. An example is the manufacture of ammonium sulfate. The synthesis stage of manufacture could be the same as the process used in the laboratory; but alternatively a manufacturer might make use of by-product or waste products from other production processes. Finding uses for by-products is an important factor in ensuring the sustainability of industrial processes. Laboratory scale procedures such as choosing reactants, synthesis, monitoring the reaction, separation techniques, disposal of waste and testing of purity have parallel counterparts in the industrial process.	6.	calculate the atom economy of a reaction to form a desired product from the balanced equation using the formula: $\text{atom economy} = \dfrac{\text{mass of atoms in desired product}}{\text{total mass of atoms in reactants}} \times 100$ M3b, M3c	**Ideas about Science** • Industrial processes can be modelled in the laboratory and using computers before making decisions about full scale production (IaS3)
	7.	use arithmetic computation when calculating atom economy M1a, M1c	
	8.	**explain why a particular reaction pathway is chosen to produce a specified product given appropriate data such as atom economy (if not calculated), yield, rate, equilibrium position, usefulness of by-products and evaluate the sustainability of the process**	
	9.	describe the industrial production of fertilisers as several integrated processes using a variety of raw materials and compare with laboratory syntheses. including: a) demand for fertilisers (including ammonium sulfate) is often met from more than one process b) some fertilisers are made as a by-product or waste product of another process c) process flow charts are used to summarise industrial processes and give information about raw materials, stages in the process, products, by-products and waste d) lab processes prepare chemicals in batches, industrial processes are usually continuous.	
	10.	compare the industrial production of fertilisers with laboratory syntheses of the same products	